Engineering Physics (2025)
Course code 25PY101
Unit 2: Quantum mechanics

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Unit 2 Plan

1. Introduction to QM
2. Dual nature of radiation
3. de Broglie's concept of matter waves
4. Heisenberg's uncertainty principle
5. Schrödinger's time dependent wave equation
6. Particle in a 1D box

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Applications of Schrödinger's W.E.

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m}{\hbar^2}\big(E - V(x)\big)\psi(x) = 0 \qquad \Psi(x,t) = \psi(x)e^{-j\omega t}$$

- The aim is to solve for the time independent wave function $\psi(x)$ for a given potential $V(x)$.
- The potential functions that are useful to describe commonly occurring physical systems are

$V(x)$	Name	Physical systems
0	Constant potential	Particle in free space
$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$	Infinite potential well	Ideal atom
$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$	Finite potential well	Work function
$V(x) = \begin{cases} V_0 & -\frac{a}{2} < x < \frac{a}{2} \\ 0 & \text{otherwise} \end{cases}$	Finite barrier	Nuclear fission Scanning Tunneling Micros

Particle in free space

$V(x)$: Constant potential

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0 \qquad \Psi(x,t) = \psi(x)e^{-j\omega t}$$

- Let us define the wave vector k as

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

so that the Time Independent Schrödinger Equation (TISE) is reduced to

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + k^2\psi(x) = 0$$

so that

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} = -k^2\psi(x)$$

$V(x)$: Constant potential – Solution of TISE

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} = -k^2\psi(x)$$

- The solution of above differential equation can be written as

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

- If we use the Euler's formula

$$e^{jx} = \cos x + j \sin x$$

then the solution can be written also as

$$\psi(x) = A' \cos kx + B' \sin kx$$

$V(x)$: Constant potential – Wave vector, Momentum and Energy

- All the wave vectors are allowed

$$k = -\infty \text{ to} + \infty$$

- The momentum is given by

$$p = \frac{h}{\lambda} \Rightarrow p = \hbar k \qquad [\because k = \frac{2\pi}{\lambda}]$$

- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

$V(x)$: Constant potential – Solution of TDSE

- The solution of Time Dependent part of Schrödinger Equation (TDSE) is given by

$$\phi(t) = e^{-j\omega t}, \qquad \text{where} \qquad \omega = \frac{E}{\hbar} = \frac{\hbar k^2}{2m}$$

- The total wave function is given by

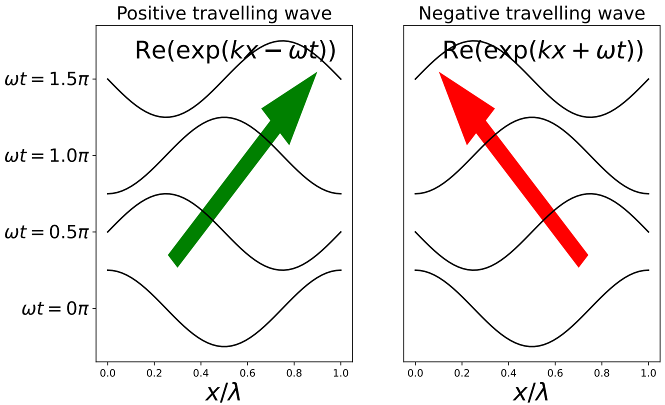
$$\Psi_k(x,t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx-\omega t}}.$$

where the subscript k is the quantum number.

$V(x)$: Constant potential – Nature of solutions

- The solutions are **traveling** waves.
- The energy E can be any positive real number and is related to the wave vector k by

$$E = \frac{\hbar^2 k^2}{2m}$$



Constant potential: problems

Problem

An electron in free space is described by a plane wave given by $\Psi(x,t) = A\exp(j(kx - \omega t))$. *If* $k = 8 \times 10^{8}\,\mathrm{m}^{-1}$ *and* $\omega = 8 \times 10^{12}\,\mathrm{rad\,s}^{-1}$, *determine*

1. *phase velocity and wavelength of the plane wave*
2. *momentum and kinetic energy (in* eV*)*

Repeat the steps for $k = -1.5 \times 10^{9}\,\mathrm{m}^{-1}$ *and* $\omega = 1.5 \times 10^{13}\,\mathrm{rad\,s}^{-1}$.

Problem

Determine the wave number, wavelength, angular frequency, and period of wavefunction that describes an electron traveling in free space at a velocity of (a) $5 \times 10^{6}\,\mathrm{cm\,s}^{-1}$.

Summary of particle in free space

- For a particle in free space, all wave vectors are allowed $k = -\infty$ to $+\infty$.
- The corresponding energy levels are given by

$$E = \frac{p^2}{2m} = \frac{\hbar^2 k^2}{2m}$$

- The time independent part of wave function is given by

$$\psi(x) = e^{jkx}$$

- The time dependent part of wave function is given by

$$\phi(t) = e^{-j\omega t}, \qquad \text{where} \qquad \omega = \frac{E}{\hbar}$$

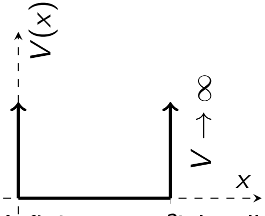
- The total wave function is given by

$$\Psi_k(x,t) = \psi_k(x)\phi_k(t) = e^{j\overline{kx-\omega t}}.$$

Particle in a 1D box

$V(x)$: Infinite potential

$$V(x) = \begin{cases} 0 & 0 < x < a \\ \infty & \text{otherwise} \end{cases}$$



Infinite potential well

- Consider a particle in a 1-dimensional infinite potential well.
- The well can be considered as a box.

Outside infinite box

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m}{\hbar^2}(E - \infty)\psi(x) = 0$$

Inside infinite box

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2mE}{\hbar^2}\psi(x) = 0$$

$V(x)$: Infinite potential well – Solution of TISE

Outside infinite box

- There is no wave function!

$$\psi(x) = 0$$

- Upon applying boundary conditions

Inside infinite box

- The wavefunction is

$$\psi(x) = A\cos kx + B\sin kx$$

Left B.C.

$$\psi(x = 0) = 0$$
$$A\cos kx = 0$$
$$\Rightarrow A = 0$$

Right B.C.

$$\psi(x = a) = 0$$
$$B\sin ka = 0$$
$$\Rightarrow \sin ka = 0$$

$V(x)$: Infinite potential well – Solution of TISE

- The equation $\sin ka = 0$ is valid if

$$ka = n\pi$$

where n is a positive integer, or $n = 1, 2, 3, \ldots$. The parameter is referred to as quantum number. In terms of the quantum number, the wave vector is given by

$$k = \frac{n\pi}{a}$$

- The prefactor B can be found by applying the **normality condition** of wave function.

$$\int_0^a |B \sin kx|^2 \, \mathrm{d}x = 1, \qquad \Rightarrow \qquad \int_0^a B^2 \sin^2 kx \, \mathrm{d}x = 1$$

$$\therefore B = \sqrt{\frac{2}{a}}$$

- Therefore, the time independent wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

- Recall that the wave vector k is related to the energy E by

$$k = \sqrt{\frac{2mE}{\hbar^2}}, \qquad \Rightarrow \qquad E = \frac{\hbar^2 k^2}{2m}$$

- Therefore, the energy is given by

$$E = E_n = \frac{n^2 \hbar^2 \pi^2}{2ma^2}$$

Problem

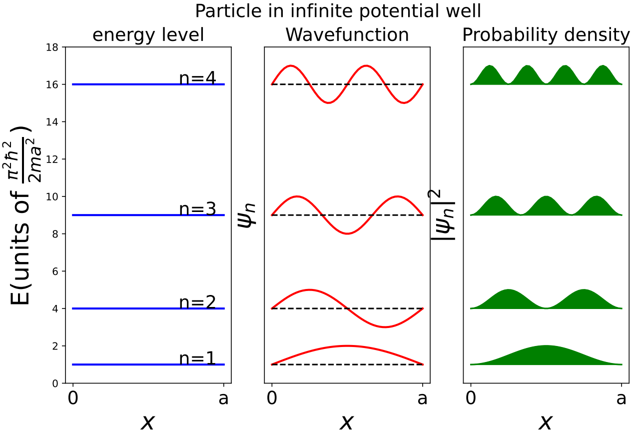
Electron in an infinite potential well of width 5 Å. Calculate the first three energy levels.

Key Insight

The energy of particle is **quantized** i.e. it can only have particular **discrete** values.

$V(x)$: Infinite potential well – Nature of solutions

- The solutions are **standing** waves.



Orthogonality condition

- Wave function associated with an energy level is normalized.
- In addition to normality, wave functions $\psi_m(x), \psi_n(x)$ corresponding to any two levels m, n of infinite potential well are orthogonal to each other i.e.

$$\int_0^a \psi_m^*(x)\psi_n(x)\,\mathrm{d}x = 0.$$

This is called the **orthogonality condition**.

Theorem

Prove the orthogonality condition for solutions of infinite potential well.

1D infinite well: problems

Problem

1. *A particle with mass of* 15 mg *is bound in a 1D infinte potential well that is* 1.2 cm *wide.*
2. *If the energy of the particle is* 15 mJ, *determine the value of* n *for that state.*
3. *What is the energy of the* $(n+1)$ *state?*
4. *Would quantum effects be observable for this particle?*

Summary of particle in infinite potential well

- For a particle in an infinite potential well, discrete wave vectors are allowed and are given by

$$k_n = \frac{n\pi}{a}.$$

- The corresponding energy levels are given by

$$E_n = \frac{\hbar^2 k^2}{2m}, \qquad \text{or} \qquad E_n = \frac{n^2\hbar^2\pi^2}{2ma^2}$$

- The time independent part of wave function is given by

$$\psi_n(x) = \sqrt{\frac{2}{a}} \sin\left(\frac{n\pi x}{a}\right)$$

- The time dependent part of wave function is given by

$$\phi_n(t) = e^{-j\omega_n t}, \qquad \text{where} \qquad \omega_n = \frac{E_n}{\hbar}$$

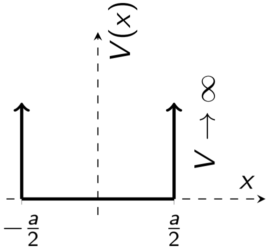
- The total wave function is given by

$$\Psi_n(x,t) = \psi_n(x)\phi_n(t),$$

Particle in a symmetric form infinite potential well

$V(x)$: Symmetric Infinite potential

$$V(x) = \begin{cases} 0 & -\frac{a}{2} \leq x \leq +\frac{a}{2} \\ \infty & \text{otherwise} \end{cases}$$



- Consider a particle in a **symmetric form** of 1-dimensional infinite potential well.
- The analysis for this potential is similar to the analysis for the asymmetric form discussed earlier.
- However, the general form of wave function can be considered as linear combination of exponential functions instead of trigonometric functions due to the symmetry of the problem.
- Thus, let the wave function inside the well be

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

$V(x)$: Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

- Upon applying left boundary condition,

$$\psi(x = -\frac{a}{2}) = 0$$

$$\Leftrightarrow A\exp\big(-j\frac{ka}{2}\big) + B\exp\big(+j\frac{ka}{2}\big) = 0$$

$$\Leftrightarrow (A+B)\cos\big(\frac{ka}{2}\big) + j(-A+B)\sin\big(\frac{ka}{2}\big) = 0$$

$$\Leftrightarrow (A+B)\cos\big(\frac{ka}{2}\big) = 0 \text{ and } (-A+B)\sin\big(\frac{ka}{2}\big) = 0$$

$$\Leftrightarrow \Big[A+B = 0 \text{ or } \cos\big(\frac{ka}{2}\big) = 0\Big] \text{ and } \Big[A-B = 0 \text{ or } \sin\big(\frac{ka}{2}\big) = 0\Big]$$

$$\Leftrightarrow \Big[A-B = 0 \text{ and } \cos\big(\frac{ka}{2}\big) = 0\Big] \text{ or } \Big[A+B = 0 \text{ and } \sin\big(\frac{ka}{2}\big) = 0\Big]$$

- Same cases arise for the right boundary condition due to **symmetry** of the problem!

$V(x)$: Symmetric Infinite potential – Solution of TISE

$$\psi(x) = Ae^{jkx} + Be^{-jkx}$$

Case (i) Even wave functions

$$A - B = 0 \quad \text{and} \quad \cos\left(\frac{ka}{2}\right) = 0$$

$$\Leftrightarrow \psi(x) = A\cos kx$$

$$\Leftrightarrow \frac{ka}{2} = \frac{\pi}{2}, \frac{3\pi}{2}, \ldots$$

$$\Leftrightarrow k = \frac{\pi}{a}, \frac{3\pi}{a}, \ldots$$

Case (ii) Odd wave functions

$$A + B = 0 \quad \text{and} \quad \sin\left(\frac{ka}{2}\right) = 0$$

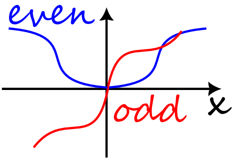
$$\Leftrightarrow \psi(x) = A\sin kx$$

$$\Leftrightarrow \frac{ka}{2} = \pi, 2\pi, \ldots$$

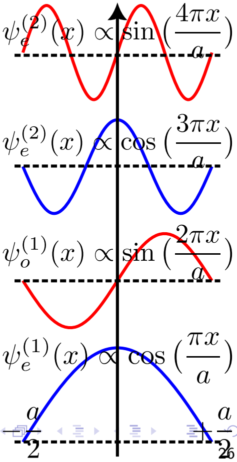
$$\Leftrightarrow k = \frac{2\pi}{a}, \frac{4\pi}{a}, \ldots$$

$V(x)$: Symmetric Infinite potential – Nature of solutions

$$f(x) := \begin{cases} \text{even} & \text{if} \quad f(-x) = f(x) \\ \text{odd} & \text{if} \quad f(-x) = -f(x) \end{cases}$$

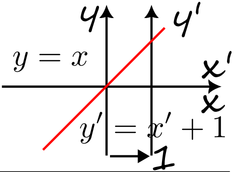


- The potential function is an even function as $V(-x) = V(x)$.
- The wave functions belonging to first case are even functions as $\cos x$ is an even function whereas the wave functions belonging to second case are odd functions as $\sin x$ is an odd function.
- There are two subfamilies of even and odd wave functions
- If we order the wave functions with increasing wave vector, then wave functions alternate between even and odd.



$V(x)$: Symmetric Infinite potential – Alternative solution

- Translation of origin by α transforms the function in new coordinate frame
$$f^{\text{new}}(x) = f^{\text{old}}(x+\alpha)$$



- The solutions to the symmetric form of infinite potential well can be obtained from the solutions to the asymmetric form by translating the coordinate frame by $\frac{a}{2}$.
$$x \to x + \frac{a}{2}$$
- The wave functions of the symmetric potential well are given by
$$\psi^{\text{sym}}(x) = \psi^{\text{asym}}(x + \frac{a}{2})$$

$$\begin{aligned}\psi^{\text{sym}}(x) &= \psi^{\text{asym}}(x + \frac{a}{2}) \\ &= \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}\overline{x + \frac{a}{2}}\right) = \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi x}{a} + \frac{n\pi}{2}\right)\end{aligned}$$

$V(x)$: Symmetric Infinite potential – Alternative solution

- Upon substituting for $n = 1, 2, \ldots$, we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \sin\left(\frac{\pi x}{a} + \frac{\pi}{2}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a} + \frac{2\pi}{2}\right) & n = 2 \\ \sin\left(\frac{3\pi x}{a} + \frac{3\pi}{2}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a} + \frac{4\pi}{2}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

- Using the trigonometric relations, we have

$$\psi^{\text{sym}}(x) \propto \begin{cases} \cos\left(\frac{\pi x}{a}\right) & n = 1 \\ \sin\left(\frac{2\pi x}{a}\right) & n = 2 \\ \cos\left(\frac{3\pi x}{a}\right) & n = 3 \\ \sin\left(\frac{4\pi x}{a}\right) & n = 4 \\ \vdots & \vdots \end{cases}$$

  In some cases, prefactor of -1 can be absorbed into normalizing factor.

- Thus, the solutions are grouped into even and odd families.

Summary of particle in symmetric infinite potential well

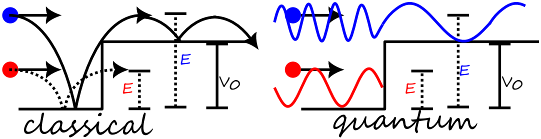
- For a particle in a symmetric infinite potential well, the allowed wave vectors are same as that of asymmetric infinite potential well.
- The corresponding energy levels are also same
- The potential well function is an even function i.e.

$$V(-x) = V(x).$$

Due to the symmetry of the potential well function, the wave functions are either **odd** or **even**.

Particle in a finite potential well

Potential energy barrier – Classical vs Quantum



- There are two regions – (I) without barrier and (II) with barrier

Region I

- Total energy E is kinetic energy T_I $$T_\mathrm{I} = E$$
- Wave vector is real $$k_\mathrm{I} = \sqrt{\frac{2mT_\mathrm{I}}{\hbar^2}}$$
- Wave function has sinusoidal form $$\psi_\mathrm{I}(x) = A \exp jk_\mathrm{I}x + B \exp -jk_\mathrm{I}x$$

Region II

- There are two cases – (i) total energy is greater than potential energy barrier and (ii) total energy is lesser than potential energy barrier.

Case (i) – $E > V_0$

- Classically, the particle overcomes the potential barrier with decreased kinetic energy.
- Quantum mechanically, the particle overcomes the potential barrier with decreased wave vector.

Region II

- Kinetic energy is reduced by V_0 $\qquad T_{\text{II}} = E - V_0$
- The wave function is given by

$$\frac{\mathrm{d}^2\psi(x)}{\mathrm{d}x^2} + \frac{2m(E - V_0)}{\hbar^2}\psi(x) = 0$$

- Wave vector is real

$$k_{\text{II}} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}}$$

- Wave vector is reduced compared to region I
- Wave function has sinusoidal form $\psi_{\text{II}}(x) = A \exp jk_{\text{II}}x + B \exp -jk_{\text{II}}x$

Curious case – $E < V_0$

- Classically, the particle **cannot** overcome the potential barrier.
- However, quantum mechanically, it is possible!
- Negative kinetic energy is allowed in quantum mechanics.

Region II

- Total energy E is less than potential barrier
- Kinetic energy is negative! $T_{II} = E - V_0 < 0$
- Wavevector is imaginary

$$k_{II} = \sqrt{\frac{2m(E - V_0)}{\hbar^2}} = j\sqrt{\frac{2m(V_0 - E)}{\hbar^2}} = j\kappa_{II}$$

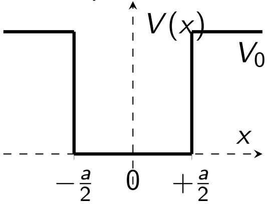
- Wave function has exponential form

$$\begin{aligned}\psi_{II}(x) &= A \exp jk_{II}x + B \exp -jk_{II}x \\ &= A \exp -\kappa_{II}x + B \exp +\kappa_{II}x\end{aligned}$$

$V(x)$: Finite potential well

$$V(x) = \begin{cases} 0 & -\frac{a}{2} < x < \frac{a}{2} \\ V_0 & \text{otherwise} \end{cases}$$

Finite potential well



- There are three regions – (I) well, (II) right of well, and (III) left of well.
- There are two cases – (i) total energy is less than potential barrier and (ii) total energy is greater than potential barrier
- Let us solve for case (i).

$V(x)$: Finite potential well – Case (i): $E < V_0$

Region I

- Kinetic energy is positive so $\psi_\text{I}(x) = A\exp(jk_\text{I}x) + B\exp(-jk_\text{I}x)$
- Apply the **symmetry condition** of even or odd wave function

$$\text{Case (1): } A = B \quad \text{or} \quad \text{Case (2): } A = -B$$

Region II

- Kinetic energy is negative so

$$\psi_\text{II}(x) = C\exp(-\kappa_\text{II}x) + D\exp(+\kappa_\text{II}x)$$

- Apply the **finiteness condition**

$$\lim_{x\to\infty} \psi_\text{II}(x) = \text{finite}$$
$$\Leftrightarrow D = 0.$$

Region III

- Kinetic energy is negative so

$$\psi_\text{III}(x) = E\exp(-\kappa_\text{III}x) + F\exp(\kappa_\text{III}x)$$

- Apply the **finiteness condition**

$$\lim_{x\to-\infty} \psi_\text{III}(x) = \text{finite}$$
$$\Leftrightarrow \quad E = 0.$$

- Due to **symmetry** of potential well $\kappa_\text{III} = \kappa_\text{II}$.

$V(x)$: Finite potential well – case (1): $A = B$

- Interface of Region I and Region II i.e. $x = \frac{\pm a}{2}$

Continuity condition of ψ

$$\psi_{\text{I}}\left(+\frac{a}{2}\right) = \psi_{\text{II}}\left(+\frac{a}{2}\right)$$
$$\Leftrightarrow A\left[\exp\left(j\frac{k_{\text{I}}a}{2}\right) + \exp\left(-j\frac{k_{\text{I}}a}{2}\right)\right]$$
$$= C\exp\left(-\frac{\kappa_{\text{II}}a}{2}\right)$$
$$\Leftrightarrow 2A\cos\frac{k_{\text{I}}a}{2} = C\exp-\frac{\kappa_{\text{II}}a}{2} \qquad (1)$$

Continuity condition of $\frac{\mathrm{d}\psi}{\mathrm{d}x}$

$$\frac{\mathrm{d}\psi_{\text{I}}}{\mathrm{d}x}\left(+\frac{a}{2}\right) = \frac{\mathrm{d}\psi_{\text{II}}}{\mathrm{d}x}\left(+\frac{a}{2}\right)$$
$$\Leftrightarrow jAk_{\text{I}}\left[\exp\left(j\frac{k_{\text{I}}a}{2}\right) - \exp\left(-j\frac{k_{\text{I}}a}{2}\right)\right]$$
$$= -C\kappa_{\text{II}}\exp-\frac{\kappa_{\text{II}}a}{2}$$
$$\Leftrightarrow 2Ak_{\text{I}}\sin\frac{k_{\text{I}}a}{2} = C\kappa_{\text{II}}\exp-\frac{\kappa_{\text{II}}a}{2} \qquad (2)$$

- Dividing (1) by (2) gives the **quantization condition** for wave vector.

$$\boxed{\tan\frac{k_{\text{I}}a}{2} = \frac{\kappa_{\text{II}}}{k_{\text{I}}}}$$

$V(x)$: Finite potential well – case (1): $A = B$

$$\tan\frac{k_{\mathrm{I}}a}{2} = \frac{\kappa_{\mathrm{II}}}{k_{\mathrm{I}}} \tag{3}$$

- This is a transcendental type of equation [1] and **cannot** be solved analytically!
- We have to employ numerical techniques.
- First let us introduce dimensionless variable z and constant z_0

$$z = \frac{k_{\mathrm{I}}a}{2} = \frac{a}{2}\sqrt{\frac{2mE}{\hbar^2}}, \qquad z_0 = \frac{a}{2}\sqrt{\frac{2mV_0}{\hbar^2}}$$

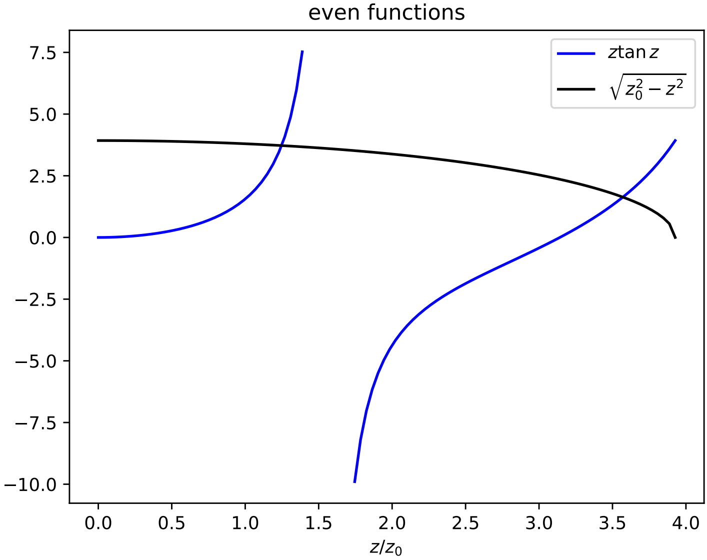
  so that (3) is reduced to

$$z\tan z = \sqrt{z_0^2 - z^2}.$$

- Now let us plot the graph of LHS and RHS of above equation. The points of intersection given you quantized values of z. From this, we can get quantized wave vectors and quantized energies.
- The depth of the well determines the number of

[1] Transcendental means beyond conventional knowledge.

$V(x)$: Finite potential well – case (1): $A = B$



$V(x)$: Finite potential well – case (1): $A = B$

- By symmetry the other interface between Region I and Region III gives the same quantization condition.
- Let us denote the quantized wave vectors as

$$k_{\mathrm{I}}^{(1)}, k_{\mathrm{I}}^{(2)}, k_{\mathrm{I}}^{(3)}, \ldots, \qquad \text{and} \qquad \kappa_{\mathrm{II}}^{(1)}, \kappa_{\mathrm{II}}^{(2)}, \kappa_{\mathrm{II}}^{(3)}, \ldots$$

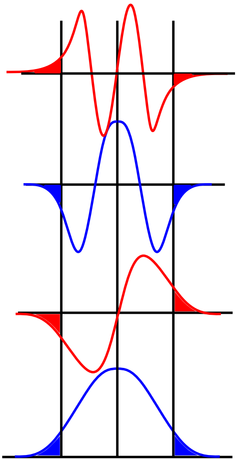
- By symmetry $F = C$.
- C can be expressed in terms of A from (1).
- Therefore, the wave functions are given by

$$\psi^{(n)}(x) = \begin{cases} C \exp\left(+\kappa_{\mathrm{II}}^{(n)} x\right) & x < -\frac{a}{2} \\ 2A \cos \frac{k_{\mathrm{I}}^{(n)} x}{2} & -\frac{a}{2} < x < +\frac{a}{2} \\ C \exp\left(-\kappa_{\mathrm{II}}^{(n)} x\right) & x > +\frac{a}{2} \end{cases}$$

- The solutions corresponding to $E < V_0$ are called **bound states**.

$V(x)$: Finite potential well – Nature of solutions

- Wave function penetrates into the finite potential barrier.
- The wave function is either odd or even function.
- The depth of the well determines the number of bound states.
- In the limiting case of very large depth, the penetration tends to zero and the number of bound states tends to infinity leading to particle in infinite potential well.



Key Insight

The wave function penetrates into the finite potential barrier. This is called **Quantum Penetration**.

- Similar analysis for the case(2): $A = -B$ gives the quantization condition as

$$\cot \frac{k_{\mathrm{I}} a}{2} = -\frac{\kappa_{\mathrm{II}}}{k_{\mathrm{I}}}$$

Problem

Prove the above quantization condition for the odd wave functions.